

AL JAHRA SULAIBIYA, Kuwait

WMO: 405870

Lat: 29.267N

Lon: 47.717E

Elev: 177

StdP: 14.60

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.8	42.7	11.3	9.8	56.7	14.9	11.8	61.2	23.4	58.9	21.5	60.3	8.1	290	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	120.2	69.3	118.5	69.1	116.6	68.7	81.2	94.4	77.7	96.1	74.3	95.5	11.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	143.2	92.1	72.5	121.2	87.5	68.1	104.0	84.9	45.1	92.5	41.2	96.8	37.9	94.7	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	24.9	22.3	19.9	DB	35.3	123.9	3.4	1.8	32.8	125.2	30.8	126.3	28.9	127.3	26.4	128.7
(5)				WB	29.9	84.1	3.8	1.6	27.1	85.3	24.9	86.3	22.7	87.2	19.9	88.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	56.9	61.9	70.8	80.5	93.0	101.2	104.0	103.4	96.4	86.1	69.4	60.0	(6)
(7)		DBStd	17.74	5.77	6.11	6.49	5.91	5.30	3.90	3.15	2.90	3.74	5.22	6.25	5.30	(7)
(8)		HDD50	15	12	2	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD65	585	256	118	18	0	0	0	0	0	0	0	25	168	(9)
(10)		CDD50	11722	226	334	646	916	1332	1535	1673	1656	1393	1118	583	310	(10)
(11)		CDD65	6817	5	31	198	466	867	1085	1208	1191	943	653	158	12	(11)
(12)		CDH74	111574	32	219	1874	5440	13825	19679	22369	21817	16100	8773	1330	116	(12)
(13)	CDH80	81519	2	45	761	2874	9592	15361	17905	17353	11836	5295	484	11	(13)	
(14)	Wind	WSAvg	9.1	7.7	8.6	8.8	8.7	9.5	12.3	12.1	10.2	8.9	7.6	7.4	7.7	(14)
(15)	Precipitation	PrecAvg	4.4	0.9	0.4	0.6	0.4	0.1	0.0	0.0	0.0	0.0	0.1	0.7	0.9	(15)
(16)		PrecMax	8.2	3.7	2.7	2.5	1.8	1.0	0.0	0.0	0.0	0.4	1.5	4.5	2.7	(16)
(17)		PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.8	0.9	0.6	0.7	0.4	0.2	0.0	0.0	0.0	0.1	0.3	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.8	85.9	101.7	106.1	116.4	121.7	122.7	121.6	117.3	108.9	95.2	81.5	(19)
MCWB			56.6	58.5	61.1	64.9	68.5	68.8	69.7	70.0	69.6	66.2	62.5	56.6	(20)	
2%		DB	73.4	79.5	93.6	100.7	112.0	119.1	120.4	119.8	113.9	105.4	90.0	77.3	(21)	
		MCWB	56.0	56.7	60.2	64.0	67.1	68.3	69.4	69.6	68.7	66.4	61.7	57.1	(22)	
5%		DB	70.2	76.2	87.7	96.6	108.9	116.6	118.3	118.1	111.6	102.4	85.5	73.6	(23)	
		MCWB	55.1	55.7	58.7	62.4	66.4	67.8	69.1	69.5	67.9	65.8	60.2	57.0	(24)	
10%		DB	67.3	73.2	83.1	93.2	105.8	114.3	116.3	116.1	109.5	99.5	81.5	70.4	(25)	
		MCWB	53.9	54.8	57.9	61.9	65.8	67.4	68.6	69.2	67.4	64.8	59.3	55.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.5	64.0	67.7	69.9	73.7	76.9	81.9	84.0	83.6	79.9	72.9	66.1	(27)
MCD B			68.6	71.6	78.3	84.1	93.0	102.6	94.1	95.9	91.7	88.7	78.9	70.8	(28)	
2%		WB	61.3	62.1	65.8	68.1	71.0	69.8	75.7	81.9	80.4	76.5	69.0	63.4	(29)	
		MCD B	67.0	70.3	75.1	82.9	94.1	114.9	101.8	95.4	93.3	85.6	76.8	70.3	(30)	
5%		WB	58.6	60.1	64.0	66.8	69.5	68.7	71.7	78.3	76.0	74.1	65.7	60.8	(31)	
		MCD B	65.7	69.0	75.2	83.5	93.4	114.1	110.6	98.6	98.0	84.7	73.8	66.8	(32)	
10%		WB	56.0	58.1	61.6	65.6	68.2	67.8	69.8	73.6	71.2	71.6	63.9	58.0	(33)	
		MCD B	63.8	68.1	77.0	84.1	96.9	113.0	114.4	104.3	99.1	82.5	73.1	66.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.9	21.1	23.4	23.0	23.6	25.5	25.5	26.9	27.1	24.9	21.2	21.2	(35)
MCDBR			22.8	24.4	30.8	27.5	26.8	28.2	27.1	28.5	28.7	28.3	26.6	25.0	(36)	
MCWBR			9.9	9.9	11.0	8.9	9.0	9.0	8.5	9.4	9.9	10.7	10.8	10.8	(37)	
5% WB		MCD BR	18.8	19.3	19.8	21.3	23.3	27.0	27.5	26.4	26.0	22.1	16.8	17.1	(38)	
		MCWBR	10.0	9.9	9.9	7.9	8.4	9.2	10.9	12.4	13.2	10.5	8.4	9.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.505	0.554	0.610	0.699	0.702	0.658	0.673	0.624	0.572	0.574	0.546	0.508	(40)	
taud		1.794	1.698	1.583	1.481	1.481	1.525	1.525	1.613	1.700	1.700	1.746	1.801	(41)		
Ebn at Noon		220	223	221	208	208	217	213	222	227	214	206	211	(42)		
Edh at Noon		58	70	83	95	96	91	91	82	73	69	61	56	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1113	1387	1745	1905	2192	2451	2355	2236	1989	1576	1158	1025	(44)	
(45)		RadStd	79	85	73	97	98	82	63	48	64	66	90	91	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.70	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	+232	+245

Nomenclature: See separate page